

Some Theoretical Questions
in Control and Dynamics
Motivated by Molecular Biology

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Part I.

Stability and Observers for Chemical Kinetics

A Question from Immunology

immune system charged with destruction and elimination of invading organisms & toxic products they produce



T-cell attacking influenza virus-infected target

“T” (thymus) lymphocytes (**type of white blood cells**)
kill virus-infected [or mutated, etc] cells
including macrophages, which ingest antigens

Kinetic Proofreading, McKeithan *PNAS*'95

in infected cells (APC = “antigen presenting cell”)
MHC molecules present foreign antigens on cell surface
(peptide fragments of foreign proteins)

must distinguish *foreign* vs *self* (autoimmune diseases!)

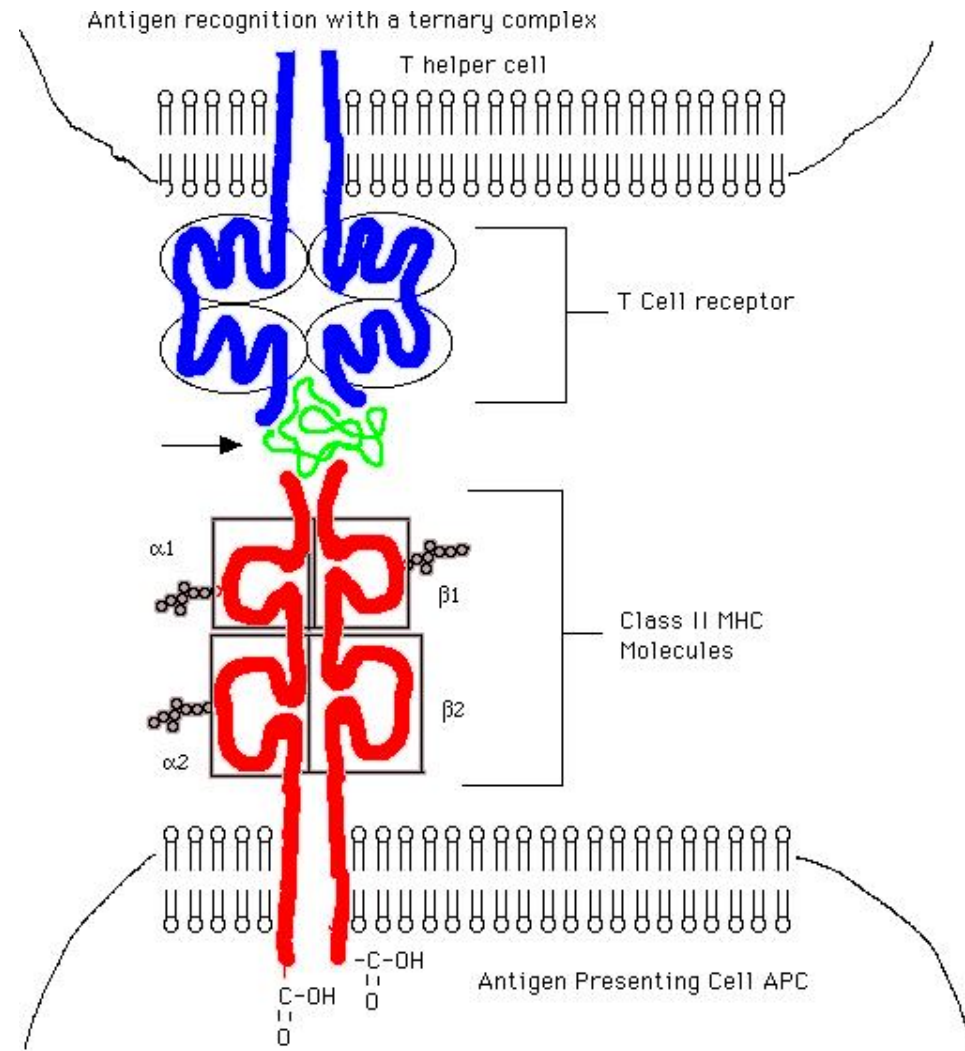
immune system distinguishes *extremely similar* antigens
(proteins differing in single aminoacid, optical isomers)

one suggestion for how selectivity is accomplished:

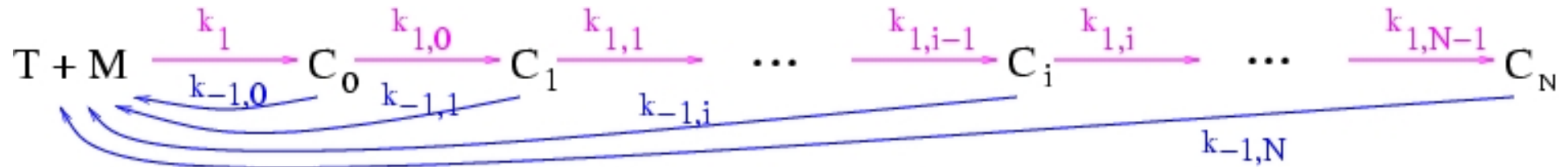
kinetic proofreading: cascade lower-selectivity steps,
with reverse error correction

“chain of modifications of T-cell receptor complex
via tyrosine phosphorylation and other reactions
↪ increased response sensitivity & selectivity”

APC & T-Cell



McKeithan's Reaction



$T(t)$ = concentration of T-cell receptor (TCR)

$M(t)$ = peptide-major histocompatibility complex (MHC)

$C_0(t)$ = initial ligand-receptor complex

$C_1(t), \dots, C_N(t)$'s = intermediate complexes

k_1 = association rate for reaction producing initial ligand-receptor complex C_0 from TCR/MHC

$k_{1,i}$ = other reaction (phosphorylation,...) rates

$k_{-1,i}$ = dissociation rates

McKeithan postulated recognition signal $\sim C_N(+\infty)$

analyzed steady states (simplifying: $k_{1,i} \equiv k_1$ & $k_{-1,i} \equiv k_{-1}$)
— in that case, easy to see $\exists$ unique steady state

Motivates Theory Questions

Carla Wofsy's talks in PIMS'99 $\rightsquigarrow$

- structure of steady states, in general?
- stability? (periodic orbits? chaos? ...?)

(unstable $\Rightarrow$ little biological significance)

EDS, IEEE TAC'01 ($\forall$ *values of kinetic constants*):

- every trajectory $\rightarrow$ steady state
- steady states unique modulo stoichiometry
- steady states form embedded submanifold, transversal to stoichiometry classes
- robustness to uncertain dynamics
- + various control problems

...contrast Belousov-Zhabotinsky, Prigogine, etc

Example: $N = 0$

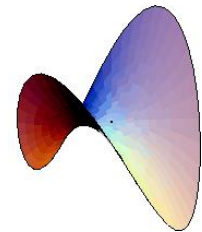


$x_i(t)$ = concentration of $T, M, C_0 \rightsquigarrow$

$$\dot{x}_1 = -k_+ x_1 x_2 + k_- x_3$$

$$\dot{x}_2 = -k_+ x_1 x_2 + k_- x_3$$

$$\dot{x}_3 = k_+ x_1 x_2 - k_- x_3,$$



steady states $\sim$ hyper.parab. $k_- x_3 - k_+ x_1 x_2 = 0$

since $d(x_1 + x_3)/dt = d(x_2 + x_3)/dt \equiv 0$, **stoichiometry classes** (lines $x_1 + x_3 \equiv c_1$, $x_2 + x_3 \equiv c_2$, $x_i > 0$) are invariant manifolds

e.g. suppose $k_+ = k_- = 1$ & take class $x_1 + x_3 \equiv 2$, $x_2 + x_3 \equiv 2$
parametrize as $(1 + \lambda, 1 + \lambda, 1 - \lambda)$, $-1 < \lambda < 1$

$\rightsquigarrow$ scalar ODE $\dot{\lambda} = -(1 + \lambda)(1 + \lambda) + (1 - \lambda) = -3\lambda - \lambda^2$

$\lambda = 0$ is globally asymptotically stable (wrt $(-1, 1)$)

corresponds to steady state $x^0 = (1, 1, 1)$

similarly for each other class $x_1 + x_3 \equiv c_1$, $x_2 + x_3 \equiv c_2$

Chemical Kinetics Formalism

systematic way to get equations,
illustrated with $T + M \rightleftharpoons C_0$:

list T, M, C_0 components in that order & introduce

labeled incidence matrix for the reaction graph $\bullet \xrightleftharpoons[k_-]{k_+} \bullet$

$$T + M \rightsquigarrow b_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad C_0 \rightsquigarrow b_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad A = \begin{pmatrix} 0 & k_- \\ k_+ & 0 \end{pmatrix}$$

$$a_{21}x^{b_1}(b_2 - b_1) + a_{12}x^{b_2}(b_1 - b_2) = k_+x_1^1x_2^1x_3^0 \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} + k_-x_1^0x_2^0x_3^1 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\rightsquigarrow \dot{x}_1 = -k_+x_1x_2 + k_-x_3, \text{ etc}$$

stoichiometric subspace $= \mathcal{D} = \text{span of } b_1 - b_2 = (1, 1, -1)$
 classes = parallel translates of $\mathcal{D}$ ($\cap$ positive orthant)

Chemical Networks

$$\dot{x} = f(x) = \sum_{i=1}^m \sum_{j=1}^m a_{ij} x_1^{b_{1j}} x_2^{b_{2j}} \dots x_n^{b_{nj}} (b_i - b_j)$$

constants a_{ij} all ≥ 0 , matrix $A = [a_{ij}]$ [block] irreducible
 $m \leq n$ ($= \#$ species = dimension of system)

each $b_j \in \mathbb{R}^n$ w/entries $b_{1j}, b_{2j}, \dots, b_{nj}$, integers ≥ 0
 $B := [b_1, b_2, \dots, b_m]$ has rank m , no row vanishes

(“zero deficiency and weakly reversible” networks
introduced by Feinberg, Horn, and Jackson in 1970s)

seen as system on $\mathbb{R}_+^n$ (forward invariant)

$\mathcal{D} := \{b_i - b_j, i \neq j\}_{\text{span}}$ invariant: $\eta \cdot (b_i - b_j) \equiv 0 \Rightarrow \eta \cdot \dot{x}(t) \equiv 0$

Basic Theorems

$E := \left\{ x \in \mathbb{R}_{\geq 0}^n \mid f(\bar{x}) = 0 \right\}$ steady states

$E_+ := E \cap \mathbb{R}_+^n$, $E_0 := E \setminus E_+$ (∂ equilibria)

positive classes: $S = (p + \mathcal{D}) \cap \mathbb{R}_{\geq 0}^n$, for each $p \in \mathbb{R}_+^n$

- E_+ is an embedded submanifold of $\mathbb{R}_+^n$,
 $\mathcal{C}^\omega$ -diffeo to $\mathbb{R}^{n-m+1}$ & $\cap$ positive classes
- $\forall x(0) \in \mathbb{R}_{\geq 0}^n$, $x(t) \rightarrow E$ as $t \rightarrow +\infty$
- $\forall S$, $\exists$ unique equilibrium $\bar{x}_S \in S \cap E_+$,
and $\bar{x}_S$ is asymptotically stable relative to S
- $\bar{x}_S$ is *globally* asymp stable rel $S \Leftrightarrow S \cap E_0 = \emptyset$
- stability robustness & feedback results

($\exists$ unique, local AS: Feinberg-Horn-Jackson)

in Kinetic Proofreading System, $S \cap E_0 = \emptyset \forall S$

some non-mass action (Michaelis-Menten) results too

Adding Outputs to Model

natural measurement functions $h : \mathbb{R}^n \rightarrow \mathbb{R}^p$:

- concentrations of some of the substances
- reaction rates (markers, fluorescence, energy released)

$\rightsquigarrow$ outputs = vectors of monomials

$$h(x) = \begin{pmatrix} x_1^{c_{11}} x_2^{c_{12}} \cdots x_n^{c_{1n}} \\ \vdots \\ x_1^{c_{p1}} x_2^{c_{p2}} \cdots x_n^{c_{pn}} \end{pmatrix},$$

where

$$C = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ & & \ddots & \\ c_{p1} & c_{p2} & \cdots & c_{pn} \end{bmatrix}$$

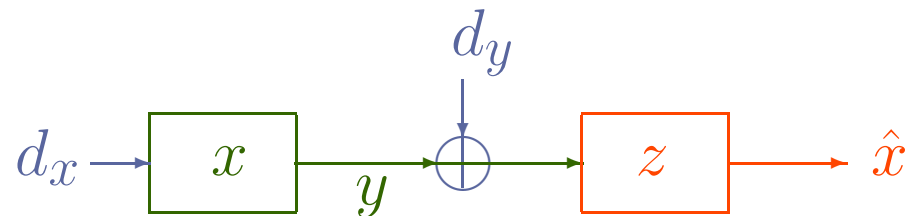
has entries integers ≥ 0 (or: 0 & reals ≥ 1)

Observers

observer for a system $\dot{x} = f(x)$, $y = h(x)$ is another system

$$\dot{z} = g(z, y), \quad \hat{x} = \hat{h}(z)$$

(typically incorporating internal model & error feedback)
that estimates internal states (e.g. chemical concentrations)
i.e.: $\forall x(0), z(0)$, composite system (x, z)



has solutions defined $\forall t \geq 0$, and $|\hat{x}(t) - x(t)| \rightarrow 0$ as $t \rightarrow +\infty$
when $\nexists$ disturbances d & “stable” behavior when $\exists d$ ’s

Kalman’s (vs. Wiener) insight for *filtering signals from noise*:
useful to have a dynamical system generating signals
 $\therefore$ observers $\rightsquigarrow$ recursive algorithms for signal filtering

Main Result on Observers

w/Madalena Chaves — NOLCOS'01, ECC'01

Theorem: $\exists$ *observer* $\iff$ “detectable”

and in that case, the following system is an observer:

$$\dot{z} = f(z) + C'(y - h(z)) \quad (\hat{x} = z)$$

- formally $\sim$ KF, but gain must be transpose of C
- difficulty is in *proving* it works

basis of proof: prove the system with inputs

$$\dot{z} = f(z) + C'(u - h(z))$$

is ISS, relative not to “ $z = 0$ and $u = 0$ ” but to equilibrium $z = x_0 = \lim_{t \rightarrow \infty} x(t)$ (always $\exists$) & input $u = h(x_0)$

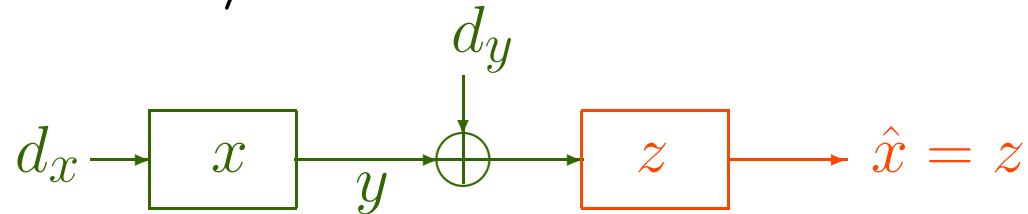
proved using an ISS-Lyapunov function ($\sim$ entropy)

now use that $x(t) \rightarrow x_0$ and cascade of ISS is ISS

bonus: automatic robustness to observation noise

Example and Simulations

illustrate with kinetic proofreading system, $N=0$: $T+M \rightleftharpoons C_0$
we test with state and/or measurement disturbances d_x, d_y



$$\dot{x}_1 = f_1(x) + d_x(t) = -k_+x_1x_2 + k_-x_3 + d_x(t)$$

$$\dot{x}_2 = f_2(x) = -k_+x_1x_2 + k_-x_3$$

$$\dot{x}_3 = f_3(x) = k_+x_1x_2 - k_-x_3$$

$$y = h(x) = \begin{pmatrix} x_1x_2^2 \\ x_1x_3 \end{pmatrix} + d_y(t)$$

$\therefore$ observer:

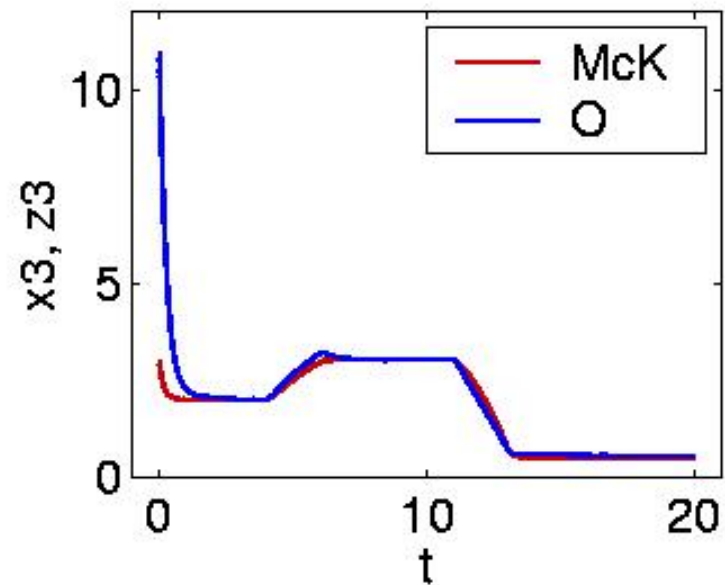
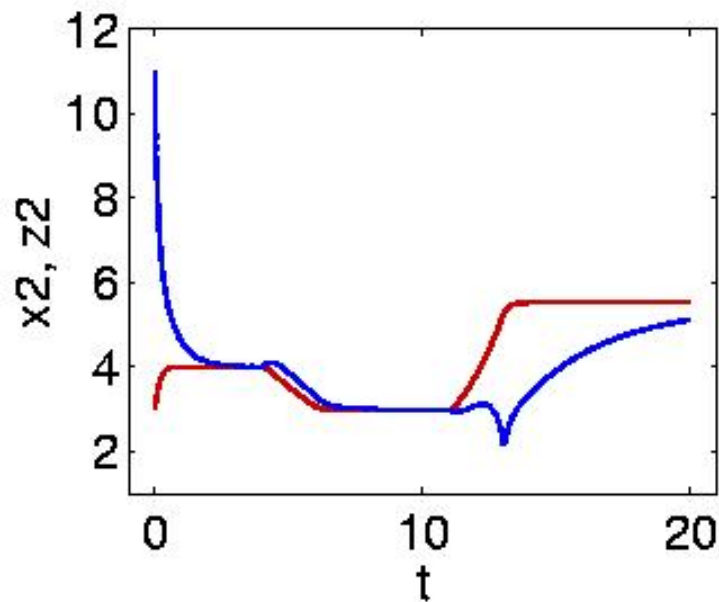
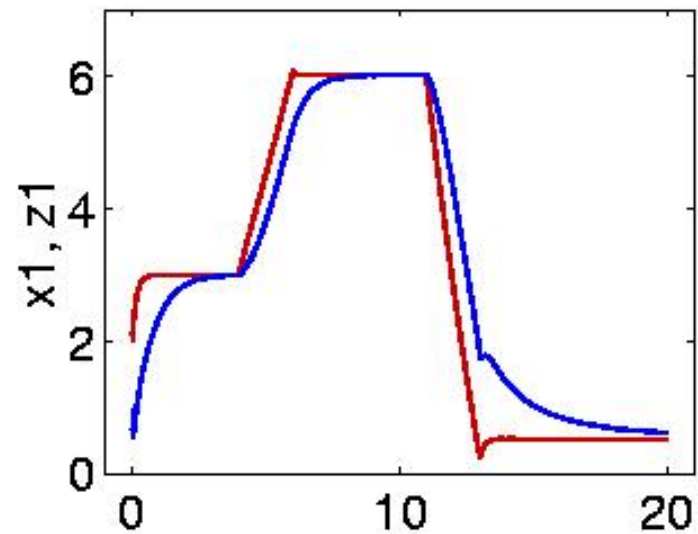
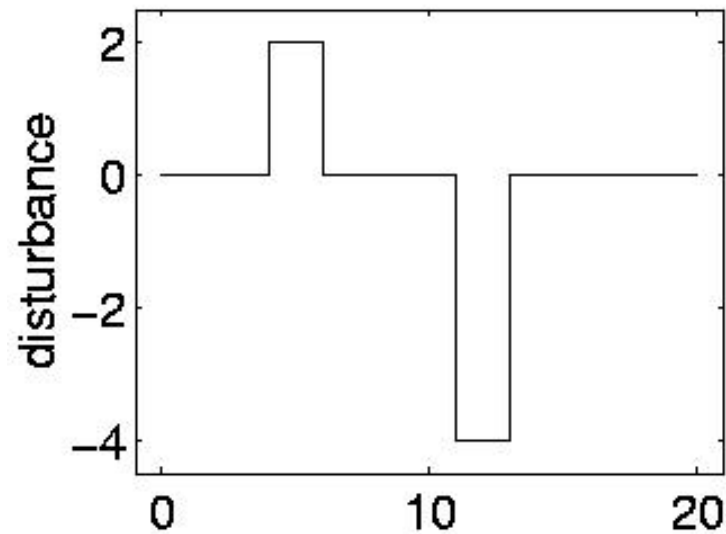
$$\dot{z}_1 = f_1(z) + (x_1x_2^2 + d_{y1}(t) - z_1z_2^2) + (x_1x_3 + d_{y2}(t) - z_1z_3)$$

$$\dot{z}_2 = f_2(z) + 2(x_1x_2^2 + d_{y1}(t) - z_1z_2^2)$$

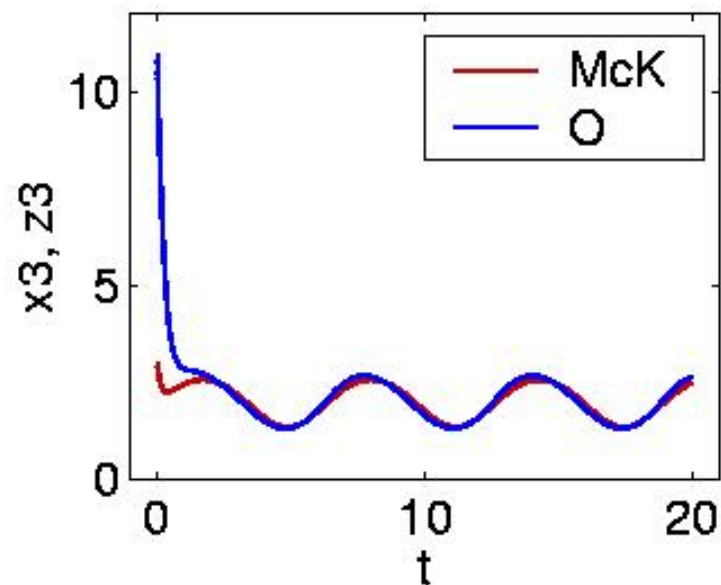
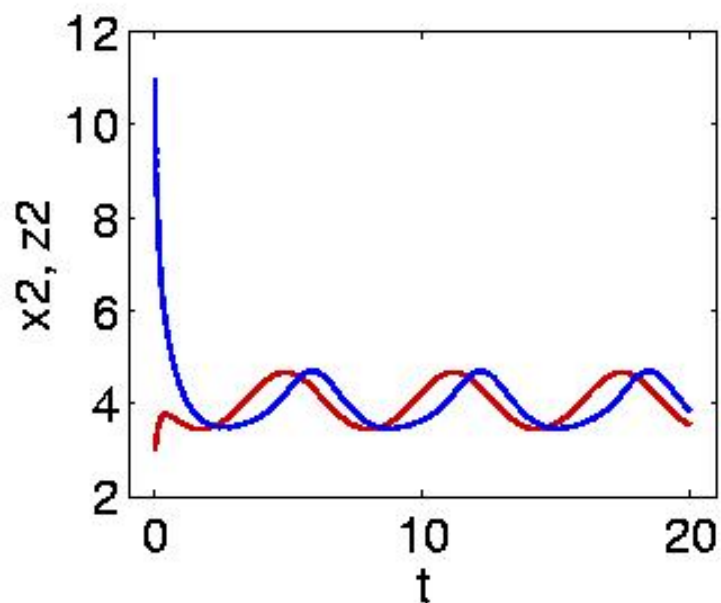
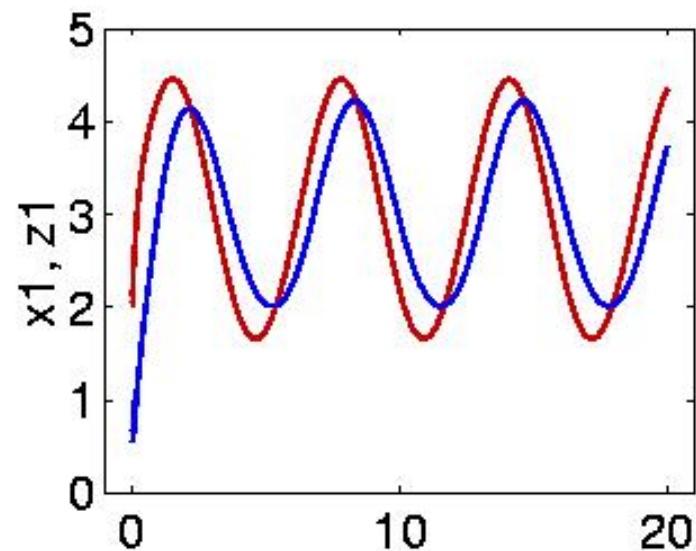
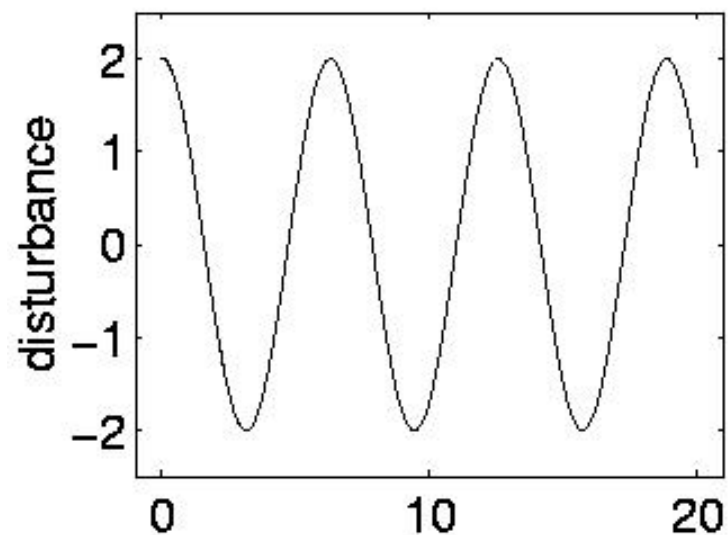
$$\dot{z}_3 = f_3(z) + (x_1x_3 + d_{y2}(t) - z_1z_3)$$

we took $k_+=0.5$, $k_-=3$; $x(0)=(2, 3, 3)'$ and $z(0)=(1, 11, 11)'$

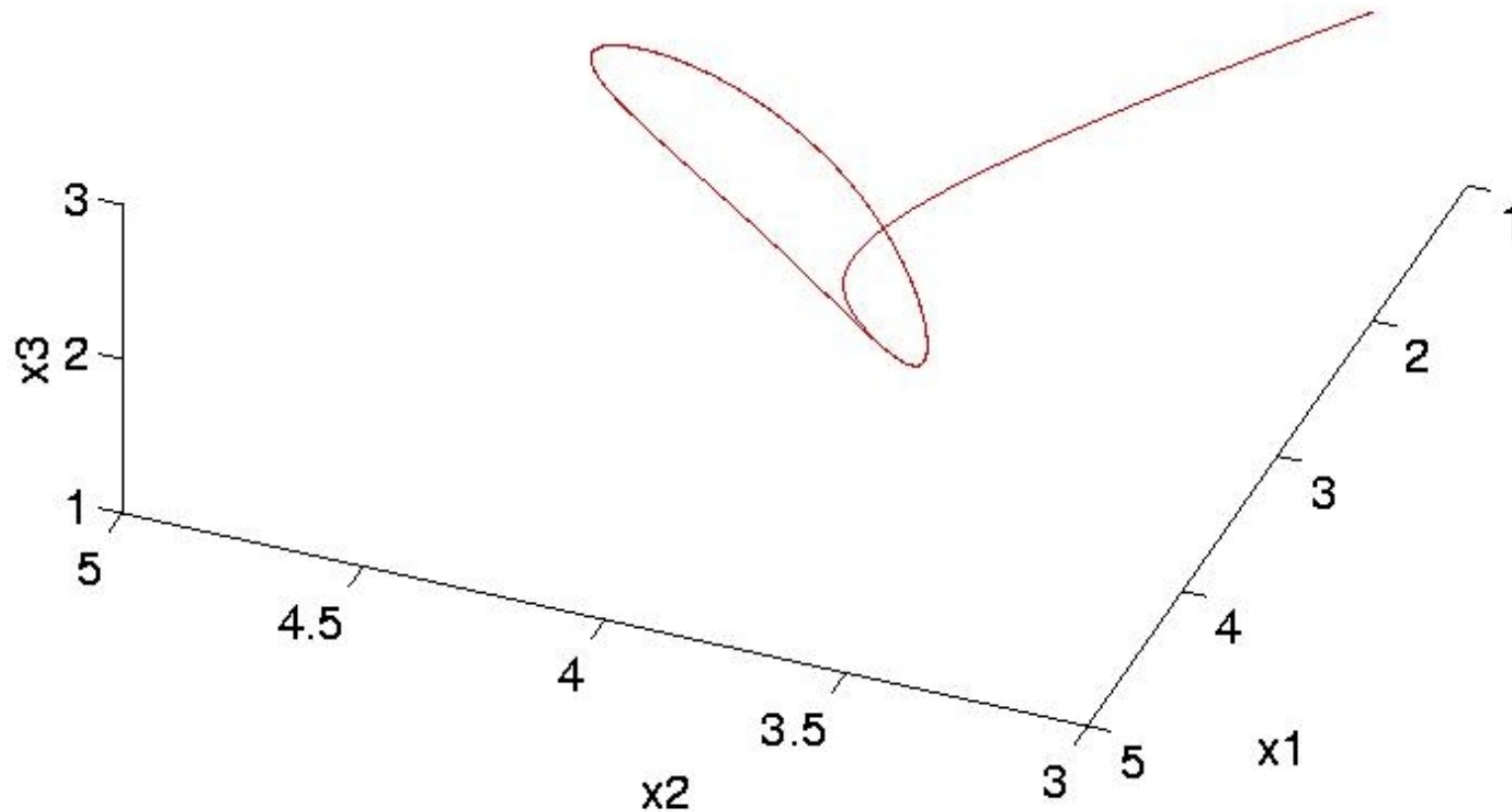
Disturbances $(d_x, d_y) = (\text{steps}, 0)$



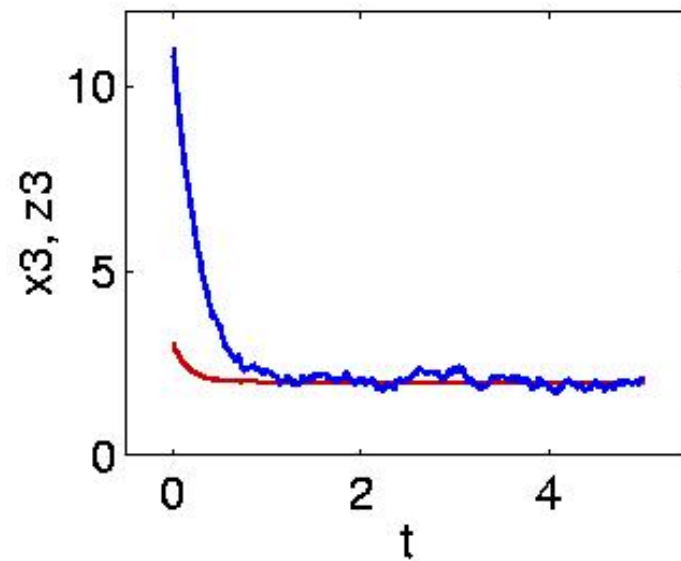
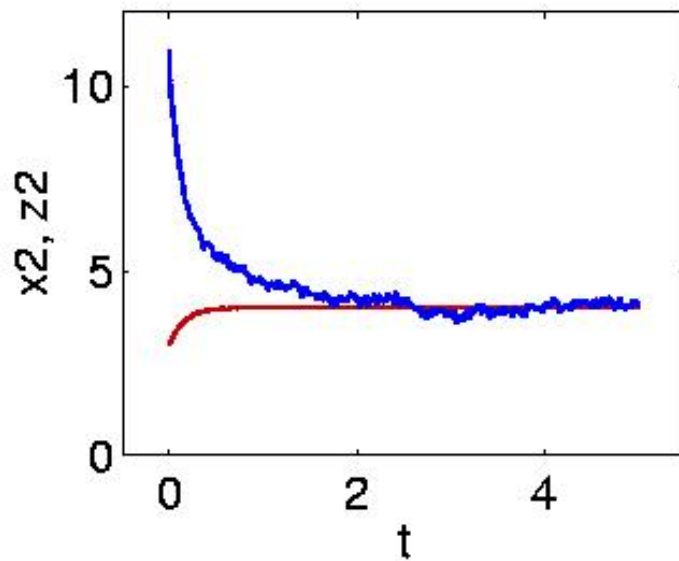
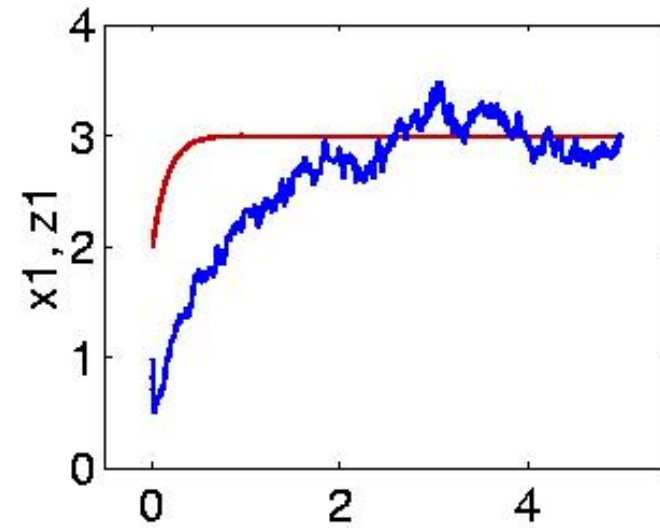
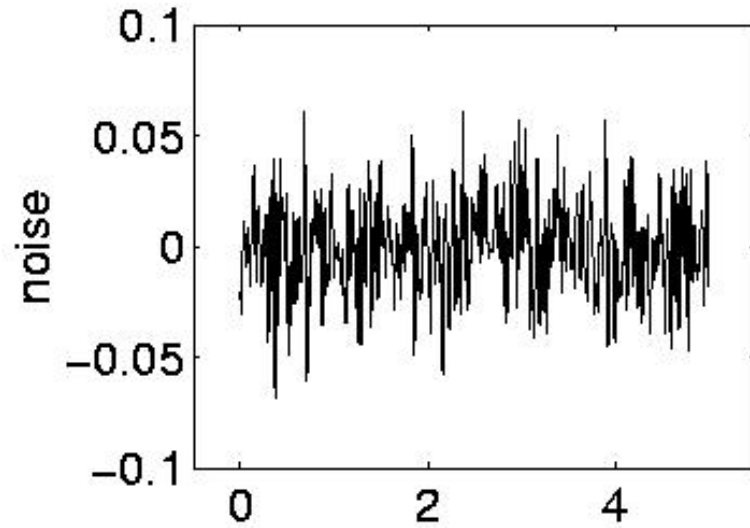
Disturbances $(d_x, d_y) = (\cos, 0)$



Cosine Disturbance, 3D Traj



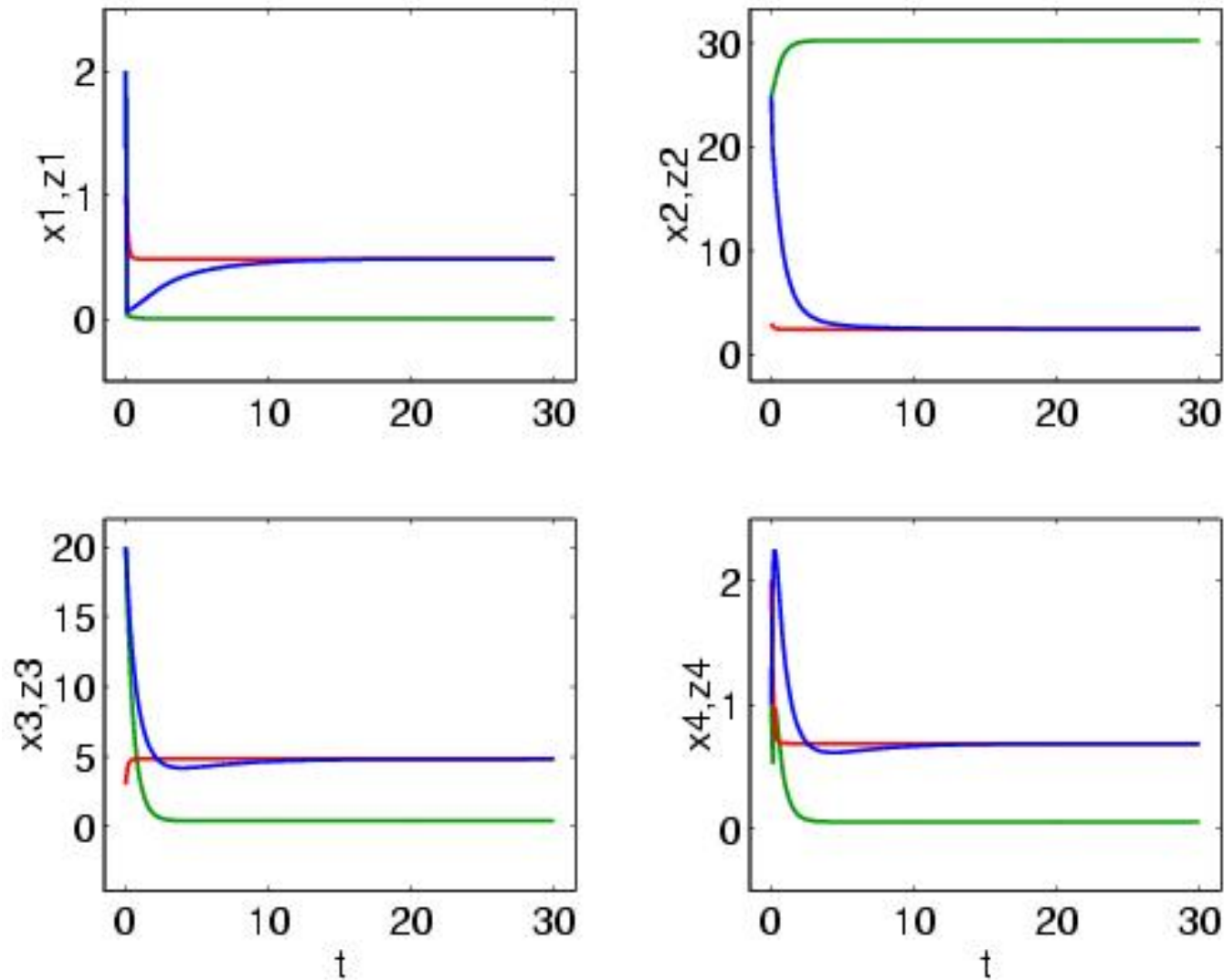
Disturbances $(d_x, d_y) = (0, \text{noise})$



Comparison with EKF

EKF may not work (or poorer convergence); e.g. $N = 1$

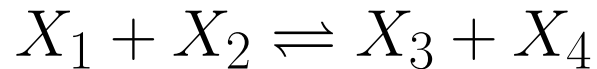
$x(0) = (1, 3, 3, 2)$, $z(0) = (2, 25, 20, 1)$, $k_1 = 6$, $k_{-1,0} = 0.5$, $k_{-1,1} = 7$, $k_{1,0} = 1$



An Experiment

Hard to do biology experiment at this time...

but chemical version done in collaboration with
Larry Romsted & Justin Mills (chemistry, RU)



X_1 = Methanol

X_2 = Acetic acid

X_3 = Water

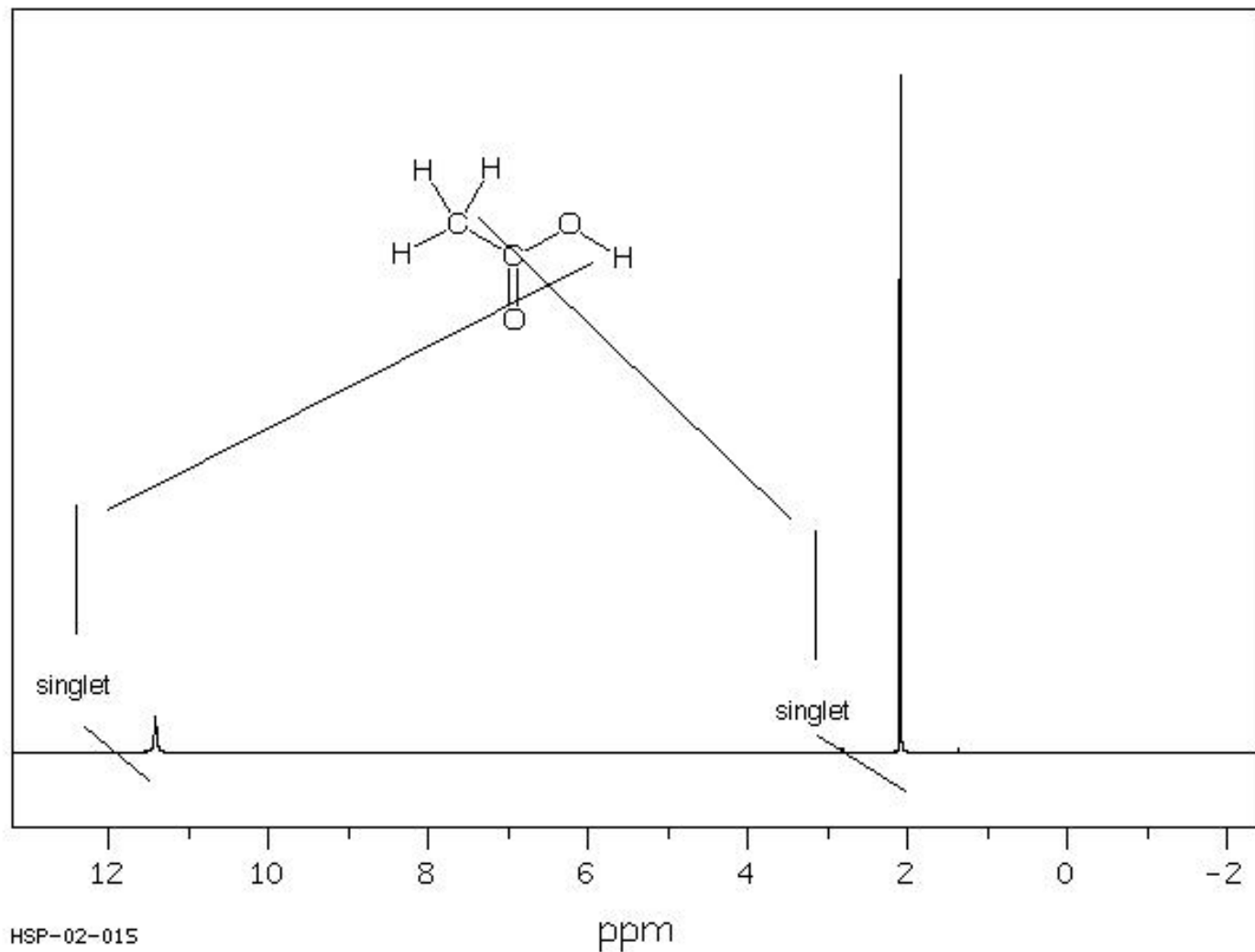
X_4 = Methyl acetate

collected data over 30 samples @ 13 min

+ sulfuric acid, trimethoxybenzene, acetonitrile- d_3

data measured by NMR spectrometry

Data Collection

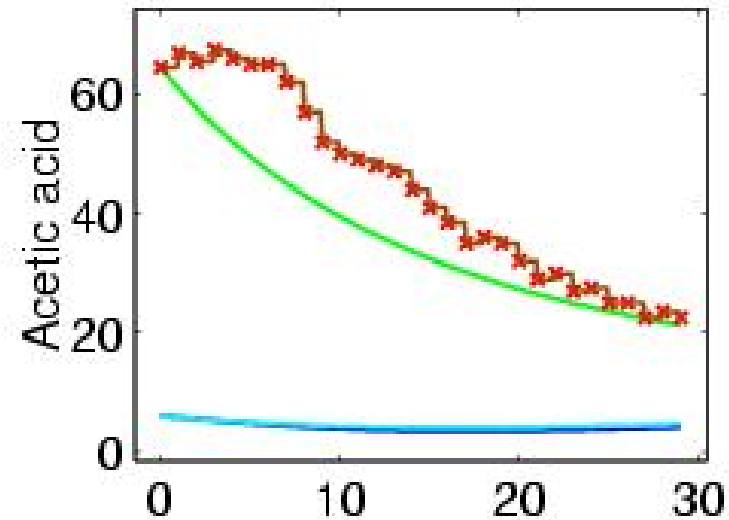
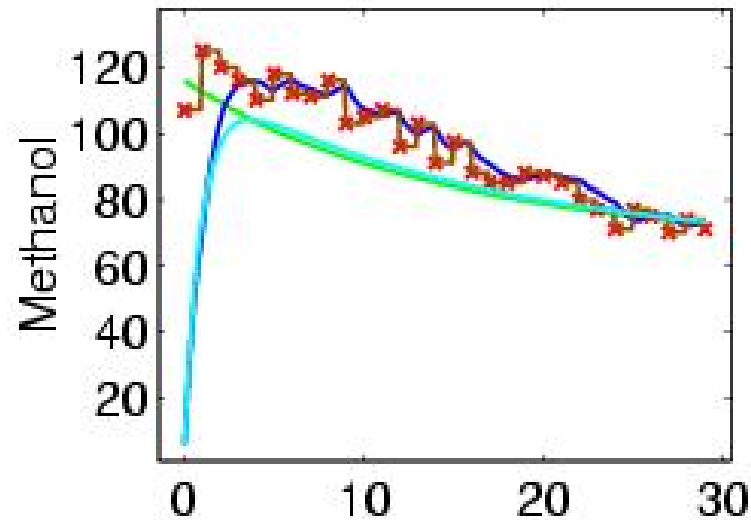


Acetic Acid Proton NMR Spectrum

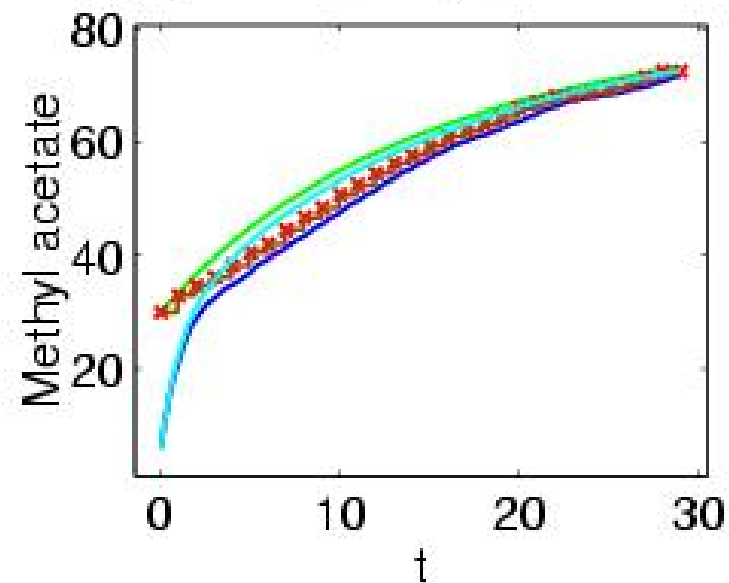
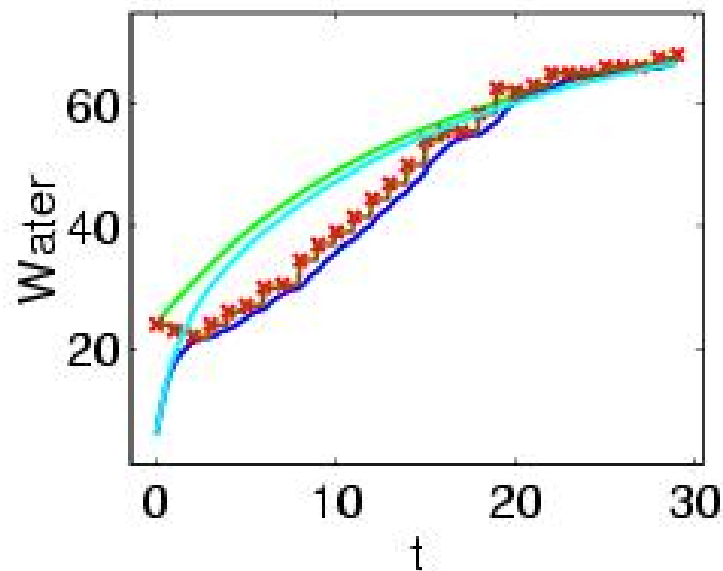
Linear Outputs

Experimental vs Theoretical

Outputs (linear): Methanol, Water, Methyl



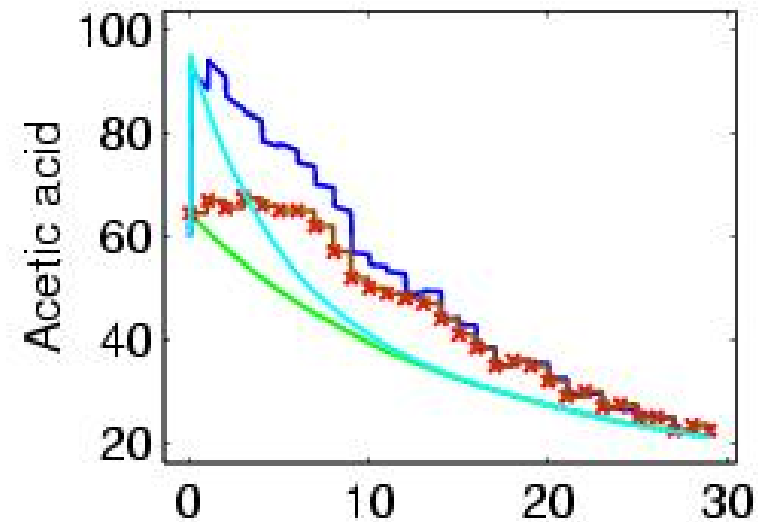
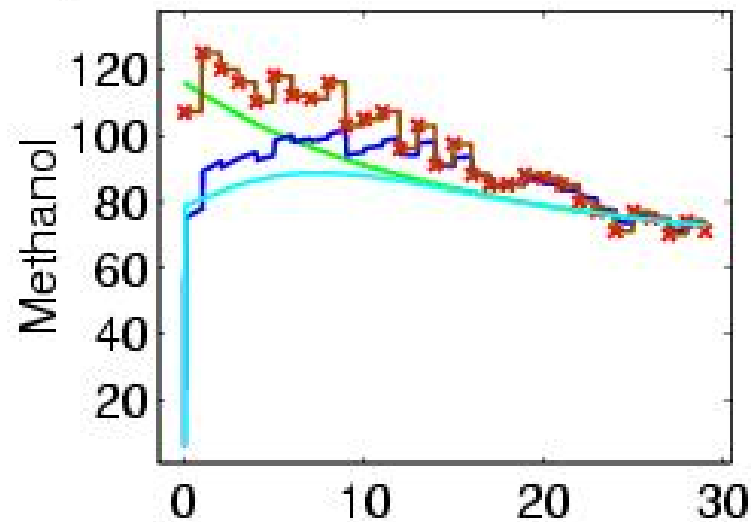
red: data & sample-hold; green: theo. model; blue: obs w/ sampling; cyan: obs w/ theo. output



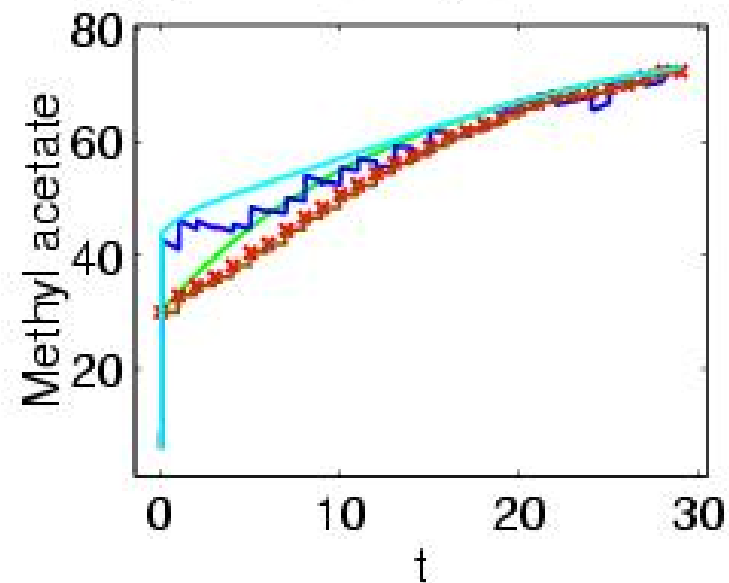
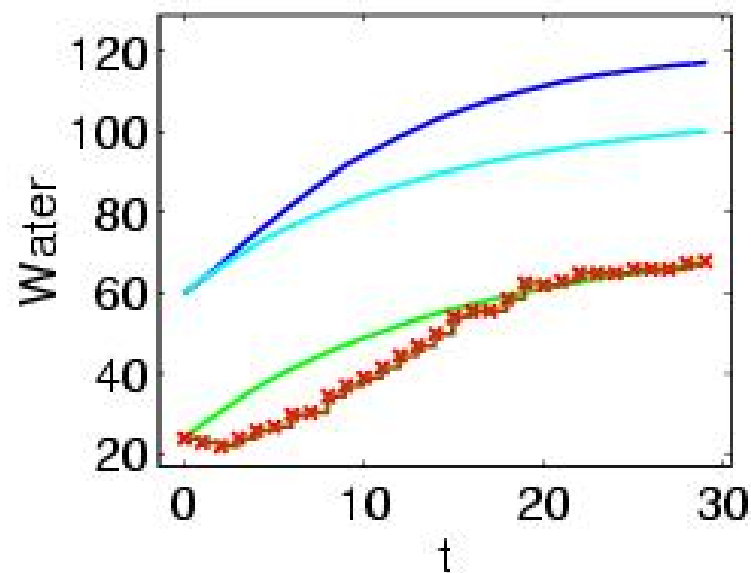
Quadratic Outputs

Experimental vs Theoretical

Outputs: Methanol * Aa, Methanol * Methyl, Methyl



red: data & sample-hold; green: theo. model; blue: obs w/ sampling; cyan: obs w/ theo. output



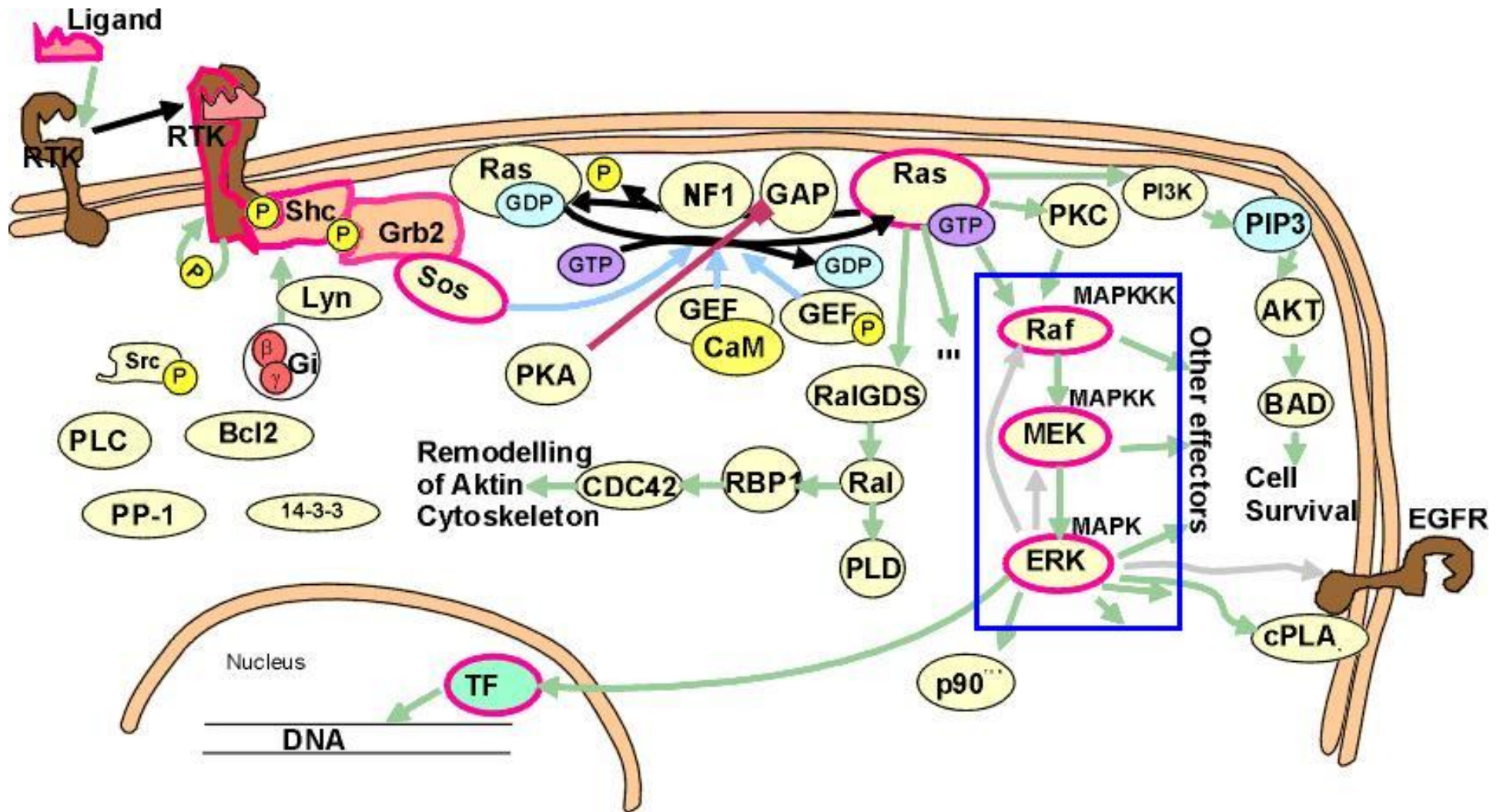
Part II.

Small-Gain Theorems, Inhibitory Feedback, and Oscillations in MAPK Cascades

MAPK (mitogen-activated protein kinase) cascades:

common signaling “submodule” in eukaryotes, involved
in cell proliferation, differentiation, movement, death

MAPK Cascades



Ras Cascade from *TRANSPATH* database

receptor tyrosine kinase as input; control of transcription factors

Common Structure

basic mechanism (with variations):
cascade of three subsystems,
each foliated into invariant manifolds

phosphorilated form drives reactions

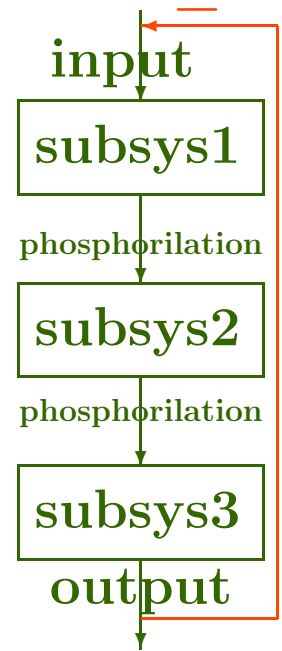
in each subsystem, phosphates attached e.g. to aminoacid tyrosine; degradation by phosphatase (phosphate-removing enzyme) produces reversibility and interesting dynamics

much current research deals with MAPK cascades

Kholodenko, European J. Biochemistry, 2000,
Shvartsman, Wiley, Lauffenburger, MIT 2000:

negative feedback (to turn-off response?)
may induce oscillations (Hopf bifurcation)

if too strong... **how strong is too strong?**



Equations

Single-phosphorilation model: write equations in terms of activated form; nondimensionalize to total “1”
state space is unit cube $[0, 1]^3 \subset \mathbb{R}^3$

$$\begin{aligned}\dot{x}_1 &= -\frac{b_1 x_1}{c_1 + x_1} + \frac{\mu}{1 + kx_3} \frac{d_1(1 - x_1)}{e_1 + (1 - x_1)} \\ \dot{x}_2 &= -\frac{b_2 x_2}{c_2 + x_2} + x_1 \frac{d_2(1 - x_2)}{e_2 + (1 - x_2)} \\ \dot{x}_3 &= -\frac{b_3 x_3}{c_3 + x_3} + x_2 \frac{d_3(1 - x_3)}{e_3 + (1 - x_3)}\end{aligned}$$

μ = external **input** to cascade (constant)

k = strength of **inhibition**

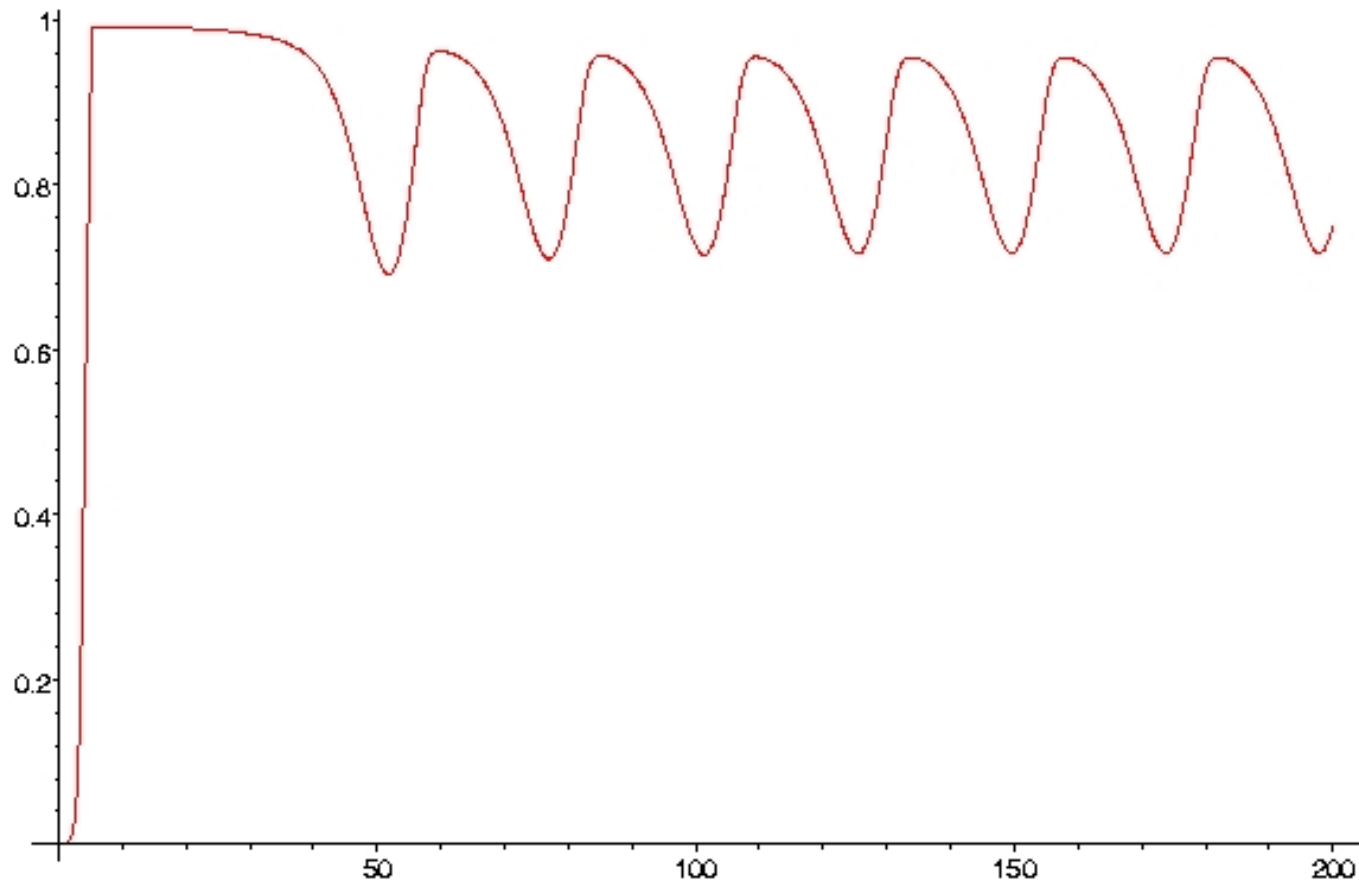
coefs from Shvartsman et. al.:

$$b_1 = c_1 = e_1 = b_2 = 0.1, c_2 = e_2 = c_3 = e_3 = 0.01,$$

$$b_3 = 0.5, d_1 = d_2 = d_3 = 1.$$

Oscillations

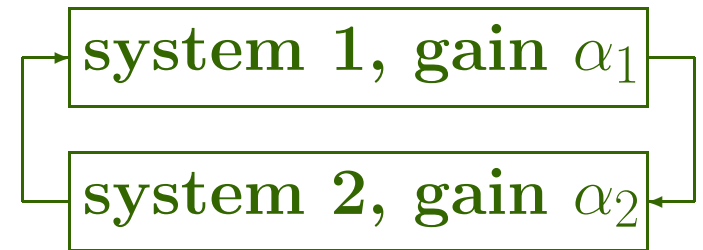
bifurcation analysis (Shvartsman) $\rightsquigarrow$ onset of oscillations:
 $k = 5.2$, $\mu = 0.3$ — graph for $x_3(t)$ with $x(0) = 0$



$\rightsquigarrow$ how to understand theoretically when non-oscillatory?

Small-Gain Principle

small gain theorems are based upon the principle that the interconnection of two stable systems is stable



provided loop  gain be less than one

i.e.: first system amplifies signals by factor α_1 , second by α_2 then round trip results in contraction by factor $\rho = \alpha_1\alpha_2 < 1$

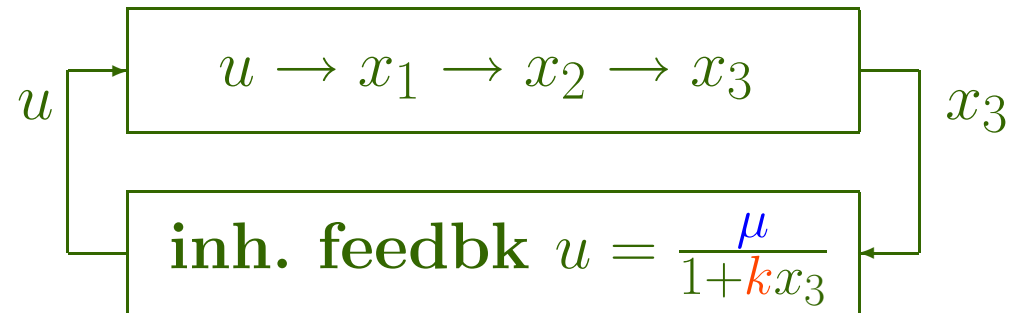
intuitively clear: signals keep getting contracted as they go around the loop, eventually converging to zero

one way to make precise: introduce norms on signal spaces in such a manner that closed-loop operator is a contraction

(theory is far more sophisticated: transient behavior, unbounded operators w/abstract “gains”, asymptotic gains)

Limitations of Usual SGT's

view MAPK as first system,
inhibitory feedback as second,
gains α_1 and $\alpha_2 = \mu k$



and ask whether small gain theorems can provide
good estimates of stable regions for parameters, ... but:
usual theorems implicitly assume equilibrium preserved

in engineering, typically negative feedback means “ $u = -kx$ ”
e.g. $\dot{x} = -x + u$, equil $x = 0$ preserved under $u = -kx$
but in biology more often $u = \frac{a}{b+kx}$, e.g.:

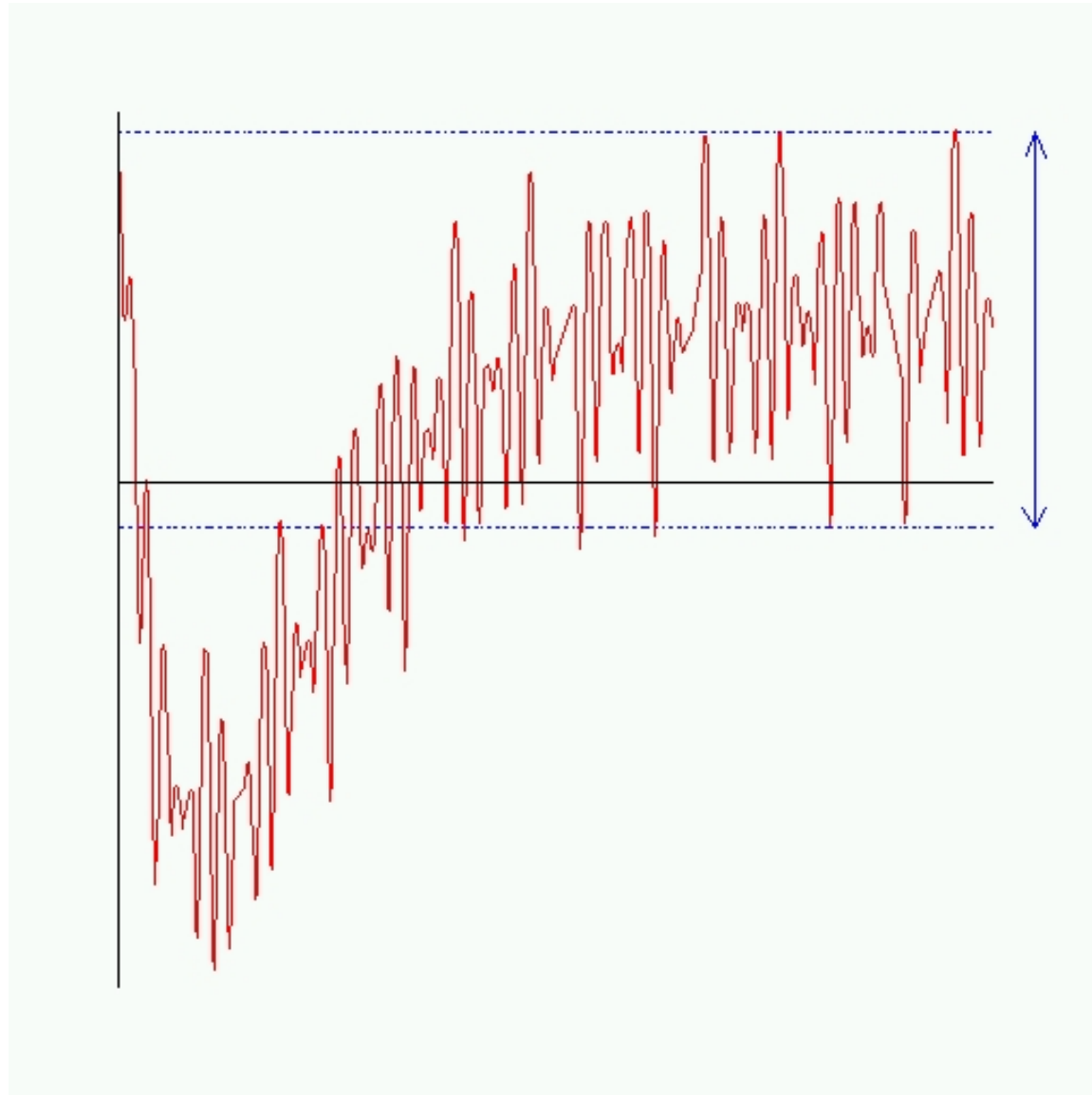
$\dot{x} = -x + \frac{4}{3+x} \Rightarrow$ new equilibrium is at $x = 1$

whose location depends on actual feedback parameters

in addition, multiple equilibria are usually of interest

$\rightsquigarrow$ new types of SGT's, new quantities associated to systems

“Asymptotic Amplitude”



$$\|\omega\|_{\mathbf{aa}} := \limsup_{s,t \rightarrow \infty} |\omega(t) - \omega(s)| \in [0, \infty]$$

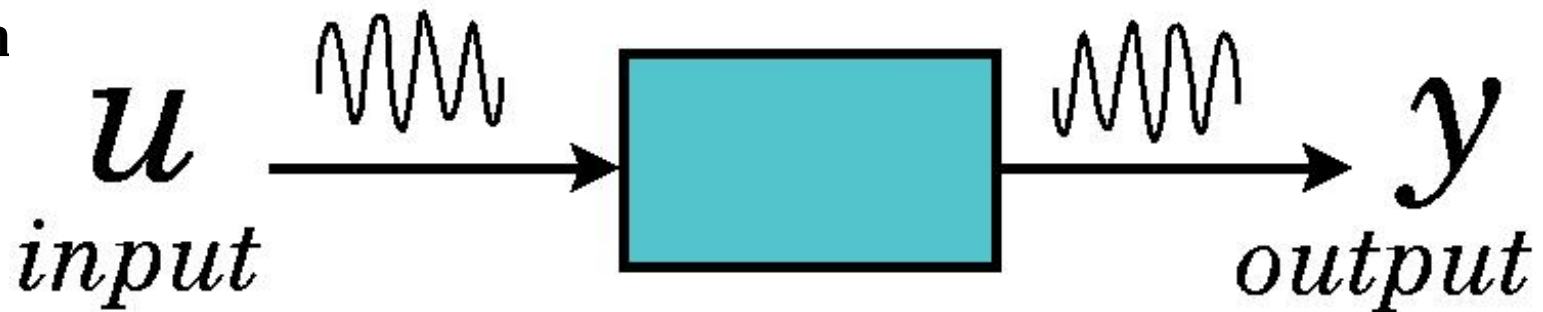
Cauchy Gains

(EDS, arXiv.org: math.OC/0112021)

$\forall$ signal $\omega : [0, +\infty) \rightarrow M$ ($M =$ Banach space, here $M = \mathbb{R}^n$)
define **asymptotic amplitude**:

$$\|\omega\|_{\mathbf{aa}} := \limsup_{s, t \rightarrow \infty} |\omega(t) - \omega(s)| \in [0, \infty]$$

system

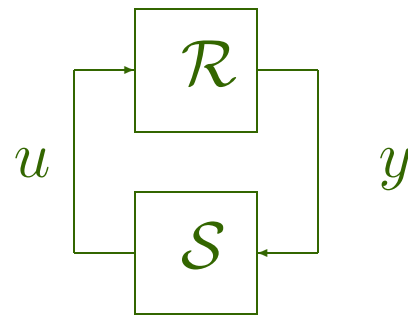


admits **Cauchy gain** $\gamma \in \mathcal{K}_\infty$ if $\|y\|_{\mathbf{aa}} \leq \gamma(\|u\|_{\mathbf{aa}}) \quad \forall u \rightsquigarrow y$

since $\|\omega\|_{\mathbf{aa}} = 0 \iff \exists \omega^\infty := \lim_{t \rightarrow \infty} \omega(t)$ (Cauchy!),

“converging input $\Rightarrow$ converging output”

Trivial SGT



closed-loop signals = intersection of graphs: $\mathcal{R} \cap \mathcal{S}^{-1}$

if $\mathcal{R}$ and $\mathcal{S}$ have Cauchy gains γ_1 and γ_2 respectively, and

$$\gamma_1(\gamma_2(r)) < r \quad \forall r > 0$$

then all bounded closed loop signals converge to equilibria:

$$\|y\|_{\mathbf{aa}} \leq \gamma_1(\gamma_2(\|y\|_{\mathbf{aa}})) < \|y\|_{\mathbf{aa}} \Rightarrow \|y\|_{\mathbf{aa}} = 0 \Rightarrow \|u\|_{\mathbf{aa}} = 0 \quad \mathbf{QED}$$

also: **incremental** gain ($\sim$ Lipschitz) condition $\Rightarrow$ **uniqueness**

with these concepts, all work goes into **estimating gains** ...

“Dissipation”-Based Estimate

observe: $\|\omega\|_{\text{aa}} = |\Omega^+[\omega]|$ (diameter)

for general state-space systems:

$$\dot{x} = f(x, u), \quad y = h(x)$$

suppose $\forall$ compact U , $\exists V = V_U$ proper, ≥ 0 s.t.:

$$x \notin Z_V := \{x \mid V(x) = 0\} \Rightarrow \nabla V(x) \cdot f(x, u) < 0 \quad (\forall u \in U)$$

and $\exists \gamma \in \mathcal{K}_\infty$ (indep of U) s.t.

$$|h(Z_{V_U})| \leq \gamma(|U|)$$

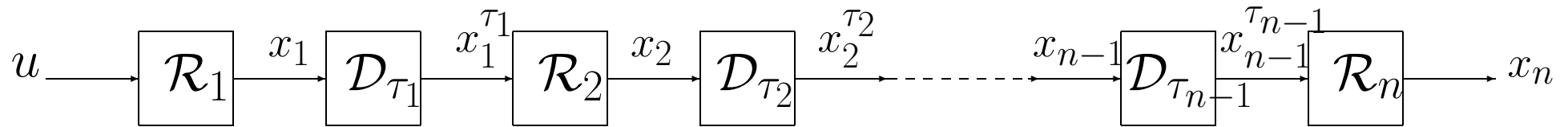
then Cauchy [and incremental] gain γ

apply to cascades of systems *for each of which* may find V 's

Apply to Cascades

even allow arbitrary inter-level delays

$$\mathcal{D}_\tau(\omega)(t) = \omega(t - \tau)$$



$$\dot{x}_1(t) = f_1(x_1(t), u(t))$$

$$\dot{x}_2(t) = f_2(x_2(t), x_1(t - \tau_1))$$

$\vdots$

$$\dot{x}_n(t) = f_n(x_n(t), x_{n-1}(t - \tau_{n-1}))$$

use gain of cascade = composition of gains

and estimate gains at each level using V 's

Even More Special Case

$$f_i(x, u) = -\alpha_i(x) + u\beta_i(x)$$

$\alpha_i \geq 0, \beta_i \geq 0$ on $[a_i, b_i]$; α_i str.increasing, β_i str.decreasing,
and $\alpha_i(a_i) = \beta_i(b_i) = 0$ ($\Rightarrow [a_i, b_i]$ invariant)

$g_i := \frac{\alpha_i}{\beta_i}$ is s.t. g_i^{-1} : constant inputs $\mapsto$ unique steady state

$\forall$ input-value set U , $V(x) :=$ distance to $g^{-1}(U)$

is good V_U ($\forall U$ connected)

and Lipschitz constants for g provide estimates on γ

observe: here, $\max (g^{-1})' = H_\infty$ -gain for linearized system
($\Rightarrow$ *local* stability) — but linear “proof” would be circular
(why should equilibria exist for closed-loop? why unique?)

in addition, new approach gives global result

Application to MAPK Model

in our application, input = inhibition $\frac{\mu}{1+kx_3}$, $x_3(t) \leq 1$,
so inputs guaranteed to be in $[\bar{u}, \infty)$ provided that $\frac{\mu}{1+k} > \bar{u}$

for each $\bar{u}$, let $\bar{x}_0 = \bar{u}$, $\bar{g}_i(x_i) := \bar{x}_{i-1}$ for $i = 1, 2, 3$

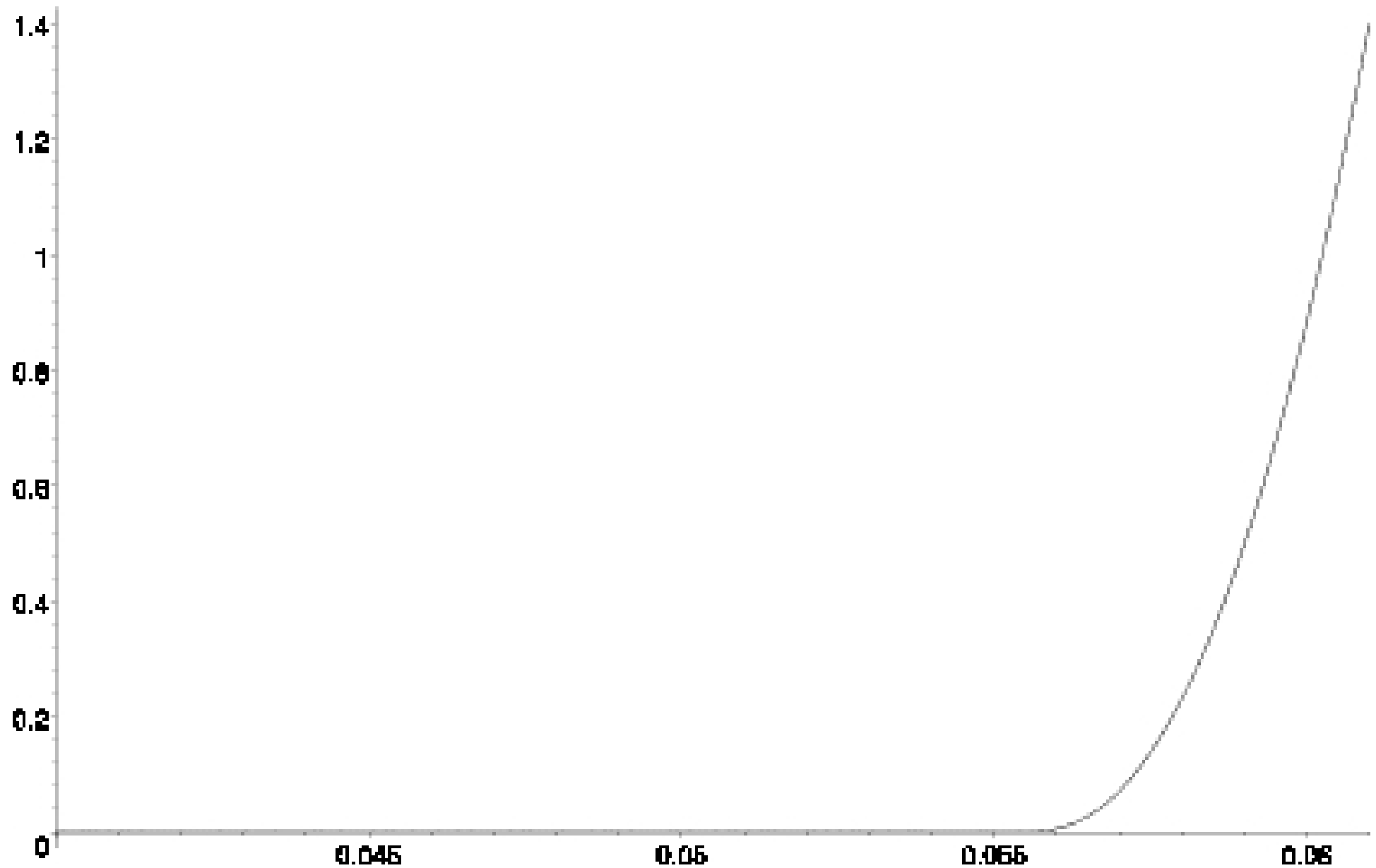
(linear) Cauchy gain of cascade upper bounded by $\theta(\bar{u}_i)^{-1}$,
where $\theta(\bar{u}) = \text{product (over } i\text{'s) of minima of } g'_i \text{ on } [\bar{x}_i, 1]$

our theory insures: inhibitory feedback $\Rightarrow$ all trajectories
converge to a (unique) equilibrium (all $\|x_i\|_{\text{aa}} = 0$) if:

$$k < \min \left\{ \frac{\theta(\bar{u})}{\mu}, \frac{\mu}{\bar{u}} - 1 \right\}$$

(small-gain condition + asking input belongs to $[\bar{u}, \infty)$)

Plot of $\theta(\bar{u})$



Estimates, MAPK Example

for each $\bar{u}$, condition

$$k < \min \left\{ \frac{\theta(\bar{u})}{\mu}, \frac{\mu}{\bar{u}} - 1 \right\}$$

$\Rightarrow$ no oscillations

note sharp transition at $\bar{u} \approx 0.055$

e.g.: ask gain be $O(1)$, so pick $\bar{u} = 0.061$, so $\theta(\bar{u}) \approx 1.4$
(linear Cauchy gain bounded by $1/\theta(\bar{u}) \approx 0.71463$)

with $\mu = 0.3$ (cf. Shvartsman et. al.) condition becomes

$$k < \min\{4.6644, 3.918\} = 3.918$$

compare: numerically, and *if* $\nexists$ delays:

$\exists$ Hopf bifurcation at $k \approx 5.1$ - so very tight even if no delays

or: secant condition (Kholodenko/Othmer/Thron/Tyson),

cf. paper: *linearized* stability if no delays and $k \leq 4.2$

Part III.

Number of Experiments Needed for Parameter Identification

Number of Experiments

ODE **system** $\dot{x} = f(x, p, u)$ involving unknown **parameters** p
(reaction constants, Hill exponents, ...)

we may apply external **inputs** $u(\cdot)$ to help excite the system
(inputs $\in$ predetermined class, e.g. constants, or pulses)

suppose we can **measure** values of some of the variables
(or of some combinations $y = h(x)$ thereof) at various times t
(measurements assumed accurate, with no observation noise)

*what is # experiments needed to obtain all the information
that is potentially available about the parameters?*

The main result is:

$2r+1$ experiments, where r is the number of parameters

best possible — also: generic $2r+1$ set OK

Motivation

central question in molbio, e.g. cell signaling pathways
simple example, just to illustrate formalism:

$$\dot{M} = \frac{E^m}{1 + E^m} - aM, \quad \dot{E} = M - bE$$

production of enzyme E & corresp. messenger RNA M
 m copies of E required to activate gene expressing E
(positive feedback) [Hill coef m not nec. integer - num fit]

state: $x = \text{vector } (M, E)$

first, **no inputs**, i.e. experiments = let system evolve,
measure $M(T)$ at end of an interval $[0, T]$ (e.g. gene arrays)
experiments parametrized only by their length T

system **parameters**: 5-dimensional vectors $p = (M_0, E_0, m, a, b)$
listing initial concentrations and constants

measurement function: $h(M, E) = M$

Variation on Example

now suppose experiments can affect degradation rate of E
e.g. $\exists$ substance which binds to (and hence neutralizes) E ,
added at concentration $u(t) \equiv q$ which we can choose
mass action kinetics controls the binding

$$\dot{M} = \frac{E^m}{1 + E^m} - aM, \quad \dot{E} = M - bE - qE$$

set of possible experiments now specified by two numbers:
 $(q, T) = (\text{neutralizer concentration, time until measure } M(T))$

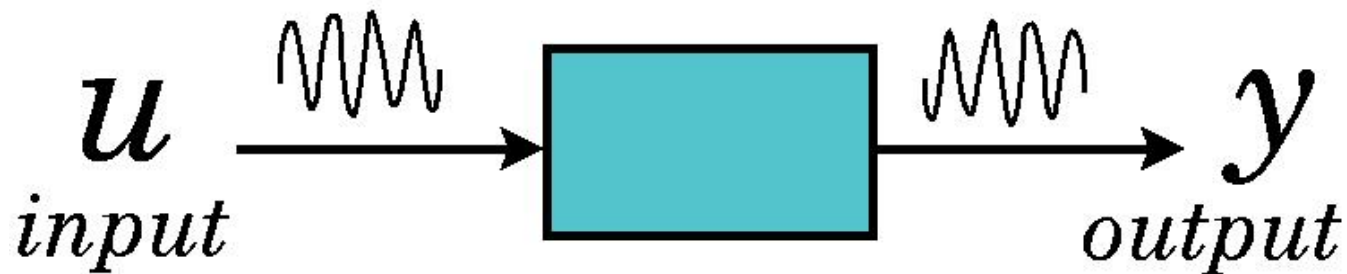
general theorem $\rightsquigarrow 2r+1 = 11$ “random experiments”
(in first case, just picking times) sufficient to distinguish

(in first, parameters not identifiable: $p = \left(1, k, \ell, \frac{k^\ell}{1+k^\ell}, \frac{1}{k}\right)$ all give $M(t) \equiv 1$,
but in second, complete identification)

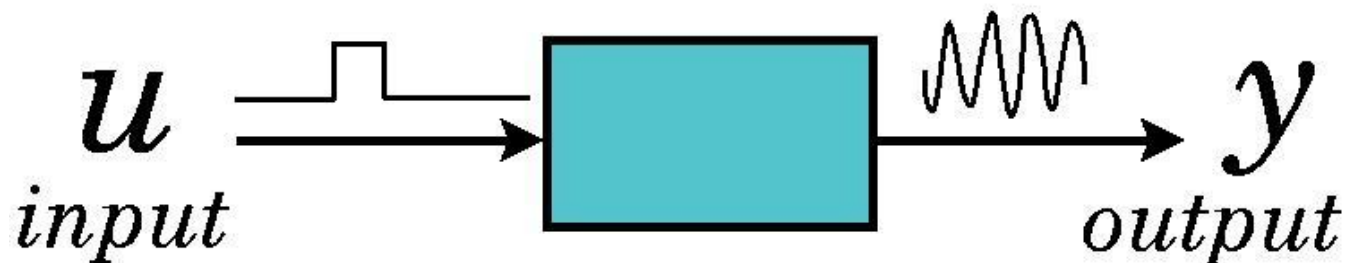
another variation: concentration is ramp $u(t) = q_0 + q_1 t$, with
 $q_0 \geq 0$ and $q_1 \geq 0$, experiments $\sim (q_0, q_1, T)$

How Different from Usual?

in identification/observation problem in systems & control
usual theorems require concatenation closure of input class;
algorithms based on e.g. sinusoidal inputs, white noise, ...



in molbio often *not* possible to apply such inputs
— e.g. nutrients, growth factors, etc. steps, pulses



so different type of question than usually asked

The Number $2r+1$

appears often in geometry and dynamical systems theory:

embedding dimension in Whitney's theorem

Takens' dimension for embedding r -dimensional attractors

Aeyels' number of samples needed for observability
of generic r -dimensional dynamical systems with no inputs

but technically our problem is quite different:
above deal with **smooth** manifolds and systems,
but we require **analytic** dependence on parameters

— main conclusion false in the more general smooth case (!)

but there are close relationships among the topics

Rigorous Definitions

analytically parametrized system is 9-tuple:

$$\Sigma = (\mathbb{X}, \mathbb{U}, \mathbb{P}, \mathbb{Q}, f, \iota, h, \mu)$$

where

- spaces of *states* $\mathbb{X}$, *input values* $\mathbb{U}$, *parameters* $\mathbb{P}$, and *input parameters* $\mathbb{Q}$ are real-analytic manifolds
- $f : \mathbb{X} \times \mathbb{P} \times \mathbb{U} \rightarrow T\mathbb{X}$, $\iota : \mathbb{P} \rightarrow \mathbb{X}$, $h : \mathbb{X} \times \mathbb{P} \times \mathbb{U} \rightarrow \mathbb{R}$, and $\mu : \mathbb{Q} \times \mathbb{R} \rightarrow \mathbb{U}$ are real-analytic maps, and
- f is a vf on $\mathbb{X}$: $f(x, p, u) \in T_x\mathbb{X}$ for each (x, p, u)

its **response** is $\beta_\Sigma : \mathbb{P} \times \mathbb{Q} \times [0, \infty) \rightarrow \mathbb{R} : (p, q, T) \mapsto y(T)$

obtained by solving IVP $\dot{x} = f(x, p, u)$, $x(0) = \iota(p)$

with input $u(t) = \mu(q, t)$,

and then evaluating $y = h(x(T))$ (assume completeness)

Distinguishability

write $\Lambda = \mathbb{Q} \times [0, \infty)$, $\lambda = (q, T)$ “experiments”

define: two parameters p_1 and p_2 **distinguishable** if:

$$(\exists \lambda \in \Lambda) \quad \beta(p_1, \lambda) \neq \beta(p_2, \lambda)$$

define: subset Λ_0 is a **universal distinguishing set** if

$$\{p_1, p_2\} \text{ distinguishable} \Rightarrow (\exists \lambda \in \Lambda_0) \quad \beta(p_1, \lambda) \neq \beta(p_2, \lambda)$$

closely related to:

- “universal inputs” in observability theory in control
- “teaching sets” in computational learning theory (Kearns)
- “experiments” in automata/sequential machines (Conway)

Summary of Results

we prove $\exists \Lambda_0$'s of cardinality $1 + 2 \dim \mathbb{P}$
and generic such sets are OK

“generic” both in topological and measure senses:

- intersection of countable family of open dense sets
- full measure

also: bound is best possible

analyticity essential in proof; false for $\mathcal{C}^\infty$

tools from stratification theory (Hironaka, Bierstone/Milman)

$\mathcal{C}^\infty$: OK only in “generic” sense (multijet transversality)

(less interesting: switches; structure; singularities; ...)

also local result: $\dim \mathbb{P}$

to do: noise!!!

Ideas

need to show $\mathcal{G}_{\beta,\rho}$ = set of universal distinguishing sets of cardinality ρ , is $\neq \emptyset$ (and also happens to be large) if $\rho = 2r+1$

study $\mathcal{B}_{\beta,\rho} := \Lambda^{(\rho)} \setminus \mathcal{G}_{\beta,\rho}$, the (“bad”) sequences:

$$\left\{ \vec{\lambda} \in \Lambda^{(\rho)} \mid (\exists p_1, p_2 \in \mathbb{P}) (p_1 \not\sim p_2 \text{ but } \forall i \beta(p_1, \lambda_i) = \beta(p_2, \lambda_i)) \right\}$$

($\not\sim$ = distinguishable, $r = \dim \mathbb{P}$)

Theorem: $\mathcal{B}_{\beta,2r+1}$ is a countable union of embedded analytic submanifolds of Λ^{2r+1} of positive codimension

($\Rightarrow \mathcal{G}_{\beta,2r+1}$ is generic in Λ^{2r+1})

— enough show $\mathcal{P}_{\beta,\rho}$ thin, since $\mathcal{B}_{\beta,\rho}$ = projection of $\mathcal{P}_{\beta,\rho} :=$

$$\left\{ \left((p_1, p_2), \vec{\lambda} \right) \in \mathbb{P}^{(2)} \times \Lambda^{(\rho)} \mid (p_1 \not\sim p_2 \text{ but } \forall i \beta(p_1, \lambda_i) = \beta(p_2, \lambda_i)) \right\}$$

Ideas, ctd.

one proves that in general, with $m = \dim \Lambda$,

$$\mathcal{P}_{\beta,\rho} = \left\{ \left((p_1, p_2), \vec{\lambda} \right) \mid (p_1 \not\sim p_2) \ \& \ \forall i \ \beta(p_1, \lambda_i) = \beta(p_2, \lambda_i) \right\}$$

is countable union of submanifolds of dimension $\leq \rho m + 2r - \rho$

$\therefore$ when $\rho = 2r+1$, $\leq (2r+1)m + 2r - (2r+1) = (2r+1)m - 1$

since $\pi_2 : \mathbb{P}^{(2)} \times \Lambda^{(\rho)} \rightarrow \Lambda^{(\rho)}$, $\pi_2(\mathcal{P}_{\beta,\rho}) = \mathcal{B}_{\beta,\rho}$, $\dim \Lambda^{2r+1} = (2r+1)m$,
 $\therefore \mathcal{B}_{\beta,2r+1} =$ countable union of submanifolds of $\text{codim} > 0$

proof of result for $\mathcal{P}_{\beta,\rho}$: consider $\text{proj } \pi_1 : \mathbb{P}^{(2)} \times \Lambda^{(\rho)} \rightarrow \mathbb{P}^{(2)}$
fibers of restriction to $\mathcal{P}_{\beta,\rho}$ are $\pi_1^{-1}(p_1, p_2) \cap \mathcal{P}_{\beta,\rho}$, the bad sets
for the given p_1, p_2 , and each has dimension $\leq \rho(m-1)$ since
included in $(\mathcal{U}(p_1, p_2))^\rho$, where $\mathcal{U}(p_1, p_2) = \{\lambda \in \Lambda \mid p_1 \sim_\lambda p_2\}$, and
each $\mathcal{U}(p_1, p_2)$ has $\dim \leq m-1$ ($p_1 \not\sim p_2$ since in $\mathcal{P}_{\beta,\rho}$)

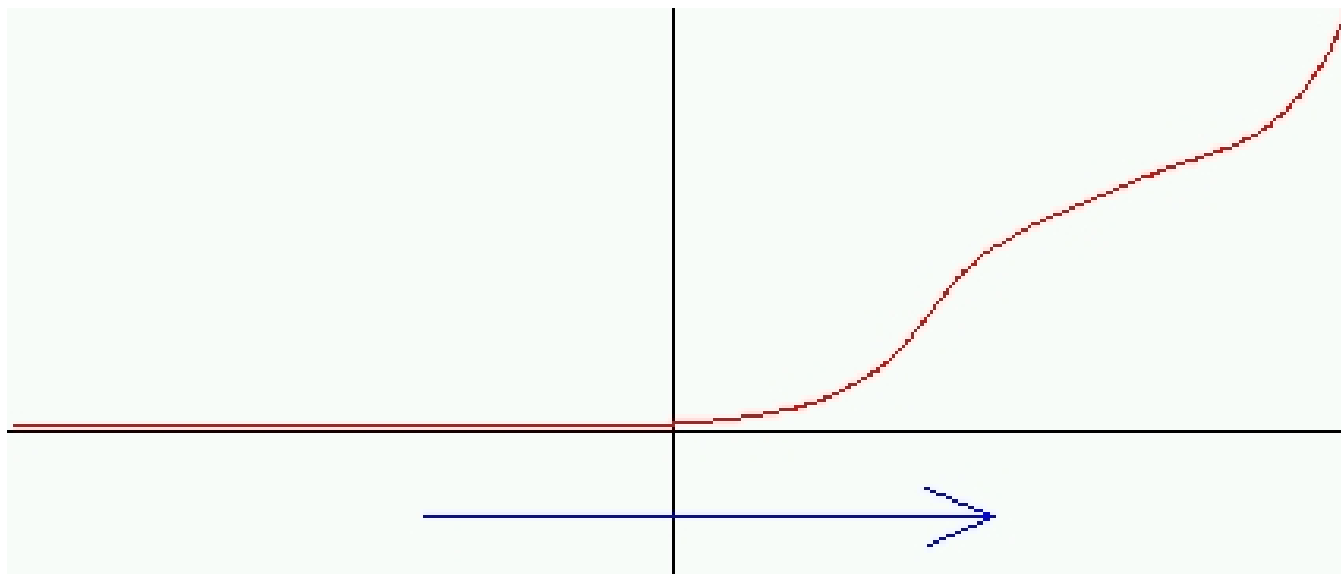
so add dimensions: $2r + \rho(m-1) = \rho m + 2r - \rho$

(analyticity used here as well as in “dimension algebra”)

False for $\mathcal{C}^\infty$

in smooth case, $2r+1$ result not true:

system $\dot{x} = 1$, $h(x) := \gamma(x)$, where γ 1-1 on $[0, \infty)$, else $\equiv 0$



is s.t. states indist. by finitely many experiments (time)
– also $\exists$ counterexample w/compact state space

(genericity result in Whitney topo via jet transversality)

Summary

- **Immunology Model $\rightsquigarrow$ Chemical Kinetics**
 - global stability relative to invariant manifolds
 - classes of systems s.t. observer theory possible
 - ... cascades: more complex systems, reaction-diffusion,
- **MAPK Cascades $\rightsquigarrow$ Multi-Equilibrium Stability**
 - new notions of gain, new small-gain theorems
 - ... develop interconnections, gain estimates, ...
- **Gene & Protein Nets $\rightsquigarrow$ Parameter Identification**
 - number of experiments
 - ... structure determination, noise, ...

documentation on this and other work available online:

<http://www.math.rutgers.edu/~sontag>